What does this get us?

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The operational semantics, and its implementations, are sound with respect to the axiomatic model

$$\frac{}{\mathbf{when}(\overline{c})\{s_1\};s_2 \hookrightarrow s_2 \mid (\overline{c}, s_1)}\ \textsc{Step}$$

$$\frac{s_1 \hookrightarrow s_2 \mid (\overline{c_3}, s_3) \qquad j\ \mathbf{fresh}}{\langle \overline{b_r}[(i, \overline{c_1}, s_1)], \overline{b_p}\rangle \,/\, H \rightsquigarrow \langle \overline{b_r}[(i, \overline{c_1}, s_2)], \overline{b_p} : (j, \overline{c_3}, s_3)\rangle \,/\, H[i \mathrel{+}= S_j]}\ \textsc{Spawn}$$

$$\frac{C = \{c \mid c \in \overline{c'} \ \wedge\ (\_, \overline{c'}, \_) \in \overline{b_r} \cup \overline{b_{p1}}\} \qquad C \cap \overline{c} = \emptyset}{\langle \overline{b_r}, \overline{b_{p1}} : (i, \overline{c}, s) : \overline{b_{p2}}\rangle \,/\, H \rightsquigarrow \langle \overline{b_r} : (i, \overline{c}, s), \overline{b_{p1}} : \overline{b_{p2}}\rangle \,/\, H[i \mathrel{+}= R_i][\overline{c} \mathrel{+}= R_i]}\ \textsc{Run}$$

$$\frac{}{\langle \overline{b_r}[(i, \overline{c}, \mathbf{done})], \overline{b_p}\rangle \,/\, H \rightsquigarrow \langle \overline{b_r}[\epsilon], \overline{b_p}\rangle \,/\, H[i \mathrel{+}= C_i][\overline{c} \mathrel{+}= C_i]}\ \textsc{Complete}$$

Figure 9 Semantics of the core BoC calculus with histories.

$$\frac{}{i \vdash_b \epsilon}\ \text{WFB-}\textsc{Empty} \qquad \frac{}{i \vdash_b R_i : \overline{S} : (C_i?)}\ \text{WFB-NE}\textsc{mpty}$$

$$\frac{}{\vdash_c \epsilon}\ \text{WFC-}\textsc{Empty} \qquad \frac{}{\vdash_c R_i}\ \text{WFC-}\textsc{Single} \qquad \frac{\vdash_c \overline{E}}{\vdash_c R_i : C_i : \overline{e}}\ \text{WFC-}\textsc{Pair}$$

Figure 10 Well-formedness rules for behaviour and cown histories

4.3 Soundness of Core Calculus According to the Axiomatic Model

In the previous section, we showed that a well-formed history translates to a valid execution in the axiomatic model. In this section we show that the semantics is sound according the axiomatic model by proving that it always produces a well-formed history.

In addition to general well-formedness, the history of an execution is going to match the specific configuration that it was produced together with.

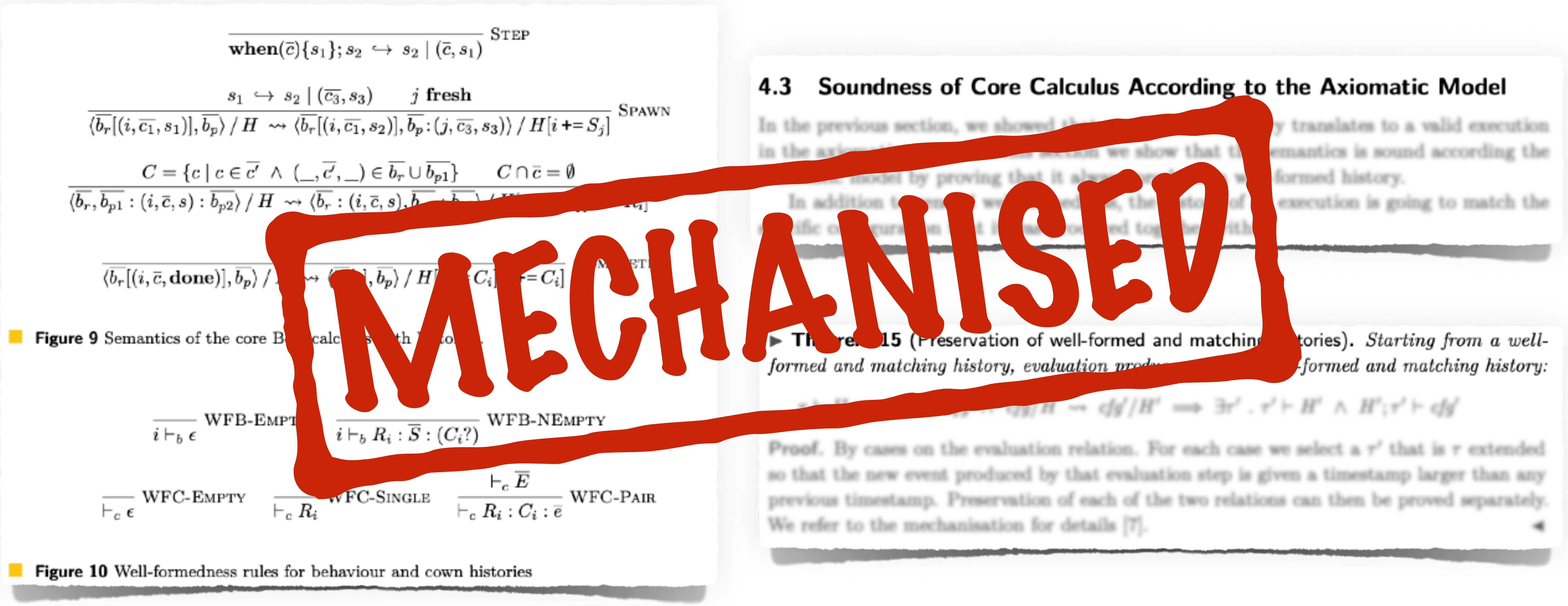
▶ **Theorem 15** (Preservation of well-formed and matching histories). *Starting from a well-formed and matching history, evaluation produces another well-formed and matching history:*

$$\tau \vdash H \ \wedge\ H; \tau \vdash cfg \ \wedge\ cfg/H \rightsquigarrow cfg'/H' \implies \exists \tau' \,.\, \tau' \vdash H' \ \wedge\ H'; \tau' \vdash cfg'$$

Proof. By cases on the evaluation relation. For each case we select a τ' that is τ extended so that the new event produced by that evaluation step is given a timestamp larger than any previous timestamp. Preservation of each of the two relations can then be proved separately. We refer to the mechanisation for details [7]. ◀

MECHANISED

What does this get us?



The operational semantics, and its implementations, are sound with respect to the axiomatic model

What does this get us?

```
when(src) { A src.balance += 10; }
when(dst) { B dst.balance += 10; }
```

```
when(dst) { C print(dst.balance) }
when(src) { D print(src.balance) }
```

	src=0, dst=0	src=0, dst=10	src=10, dst=0	src=10, dst=10
Order	C B D A	B C D A	A C B D	A B C D
	C D A B	B D A C	A C D B	A B D C
	C D B A	B D C A	A D C B	A D B C
	D A C B	D A B C	C A B D	B A C D
	D C A B	D A B C	C A D B	B A D C
	D C B A	D B C A	C B A D	B C A D
Permitted	✓	⚠	✓	✓